

Aim/Overview of the practical: b) Perform REST API testing using Postman to validate RESTful web services by executing HTTP methods such as GET, POST, PUT, and DELETE, and by verifying responses, status codes, headers, and payload data.

Objective: To perform REST API testing using Postman by validating GET, POST, PUT, and DELETE requests, and verifying HTTP status codes, headers, response payloads, and overall API functionality.

Test Cases

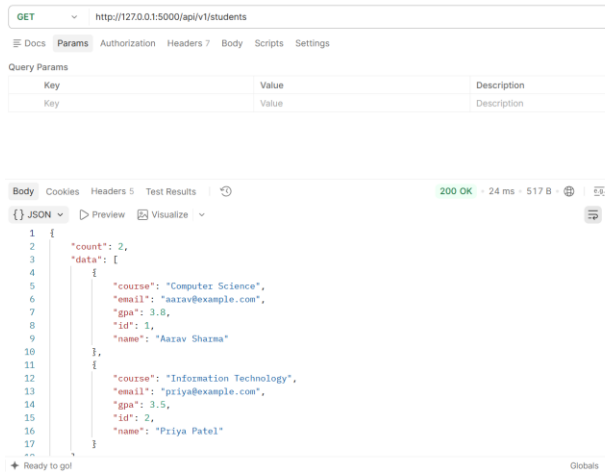
Test ID	Test Scenario / Description	HTTP Method	Endpoint	Headers / Params	Expected Status Code	Expected Result	Actual Result	Status
TC01	Fetch single existing student by valid ID	GET	/api/v1/students/1	None	200 OK	Returns student details for ID 1	Returned status 200 OK with student object for ID 1	Pass
TC02	Fetch student with non-existent ID	GET	/api/v1/students/999	None	404 Not Found	Error message: "Student not found"	Returned status 404 Not Found with message "Student not found"	Pass
TC03	Create student with valid token & body	POST	/api/v1/students	Auth: Bearer valid-token	201 Created	Student created with generated ID	Returned status 201 Created with generated ID 3	Pass
TC04	Create student without Authorization header	POST	/api/v1/students	None	401 Unauthorized	Error message: "Unauthorized or invalid token"	Returned status 401 Unauthorized with error message	Pass
TC05	Update student details with valid payload	PUT	/api/v1/students/1	Auth: Bearer valid-token	200 OK	Updated student object returned	Returned status 200 OK with updated GPA field	Pass
TC06	Update non-existent student	PUT	/api/v1/students/999	Auth: Bearer valid-token	404 Not Found	Message: "Student not found"	Returned status 404 Not Found with error message	Pass

TC07	Delete existing student with valid token	DELETE	/api/v1/students/2	Auth: Bearer valid-token	200 OK	Success deletion response	Returned status 200 OK and student ID 2 removed	Pass
TC08	Test unsupported HTTP method on resource	PATCH	/api/v1/students	None	405 Method Not Allowed	Returns 405 Method Not Allowed	Returned status 405 Method Not Allowed	Pass

OUTPUT

GET Method

200 OK



GET http://127.0.0.1:5000/api/v1/students

Query Params

Key	Value	Description
Key	Value	Description

Body

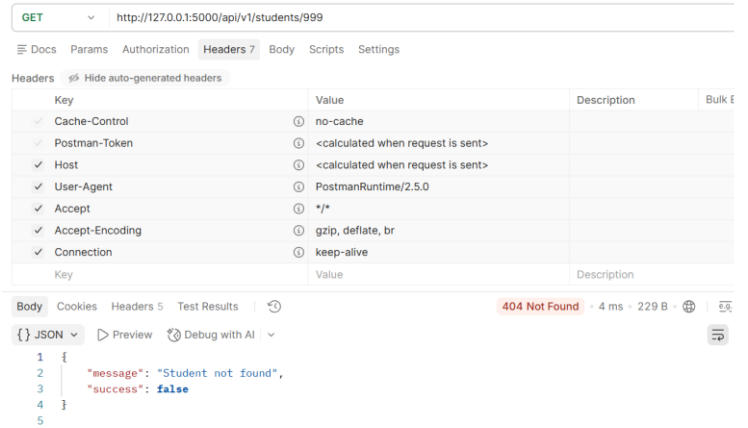
```

1 {
2   "count": 2,
3   "data": [
4     {
5       "course": "Computer Science",
6       "email": "aarav@example.com",
7       "gpa": 3.8,
8       "id": 1,
9       "name": "Aarav Sharma"
10    },
11    {
12      "course": "Information Technology",
13      "email": "priya@example.com",
14      "gpa": 3.5,
15      "id": 2,
16      "name": "Priya Patel"
17    }
18  ]
19 }

```

200 OK - 24 ms - 517 B

404 Not Found



GET http://127.0.0.1:5000/api/v1/students/999

Headers

Key	Value	Description	Bulk Edit
Cache-Control	no-cache		
Postman-Token	<calculated when request is sent>		
Host	<calculated when request is sent>		
User-Agent	PostmanRuntime/2.5.0		
Accept	*/*		
Accept-Encoding	gzip, deflate, br		
Connection	keep-alive		

Body

```

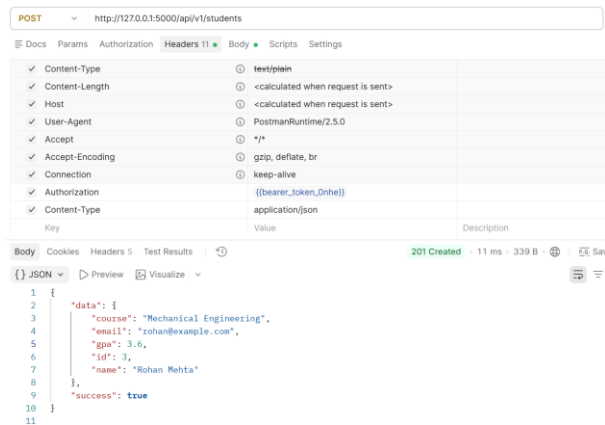
1 {
2   "message": "Student not found",
3   "success": false
4 }
5

```

404 Not Found - 4 ms - 229 B

POST Method

201 Created



POST http://127.0.0.1:5000/api/v1/students

Headers

Key	Value	Description
Content-Type	text/plain	
Content-Length	<calculated when request is sent>	
Host	<calculated when request is sent>	
User-Agent	PostmanRuntime/2.5.0	
Accept	*/*	
Accept-Encoding	gzip, deflate, br	
Connection	keep-alive	
Authorization	{bearer_token, onhell}	
Content-Type	application/json	

Body

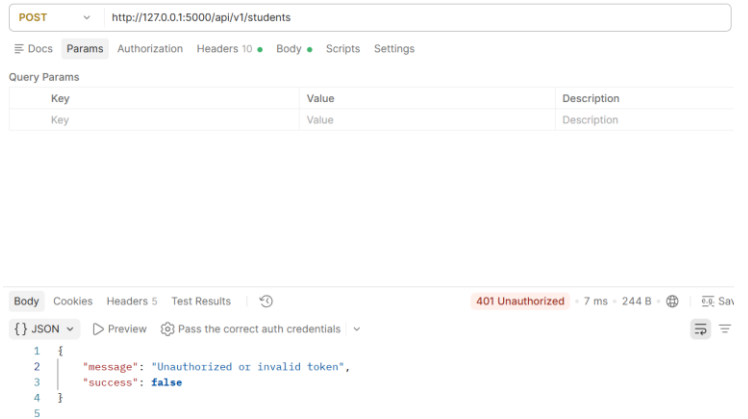
```

1 {
2   "data": {
3     "course": "Mechanical Engineering",
4     "email": "rohan@example.com",
5     "gpa": 3.6,
6     "id": 3,
7     "name": "Rohan Mehta"
8   }
9 }
10
11

```

201 Created - 11 ms - 339 B

401 Unauthorized



POST http://127.0.0.1:5000/api/v1/students

Query Params

Key	Value	Description
Key	Value	Description

Body

```

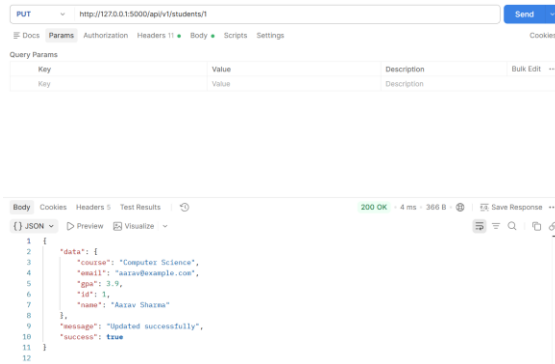
1 {
2   "message": "Unauthorized or invalid token",
3   "success": false
4 }
5

```

401 Unauthorized - 7 ms - 244 B

PUT Method

200 OK



PUT <http://127.0.0.1:5000/api/v1/students/1> Send

Auth type: No Auth

Key	Value	Description	Bulk Edit
Key	Value	Description	

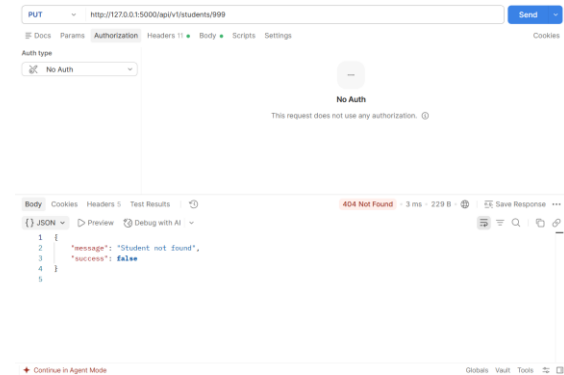
Body: 200 OK - 4 ms - 366 B

```

1 {
2   "data": {
3     "course": "Computer Science",
4     "email": "naac@example.com",
5     "gpa": 3.9,
6     "id": 1,
7     "name": "Razav Shazam"
8   },
9   "message": "Updated successfully",
10  "success": true
11 }
12

```

404 Not found



PUT <http://127.0.0.1:5000/api/v1/students/999> Send

Auth type: No Auth

No Auth

This request does not use any authorization.

Body: 404 Not Found - 3 ms - 229 B

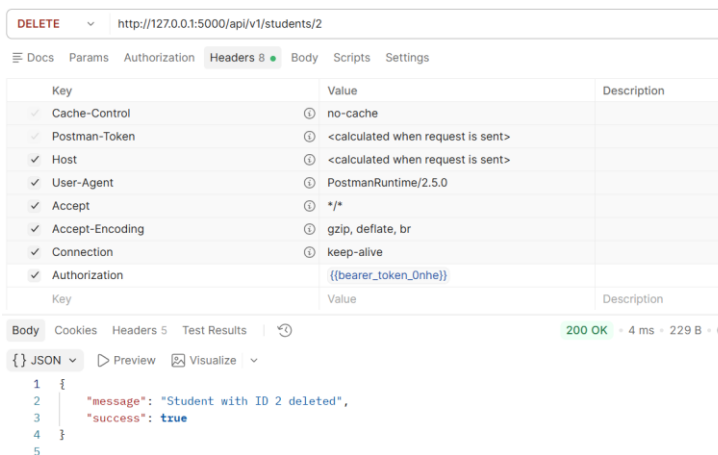
```

1 {
2   "message": "Student not found",
3   "success": false
4 }
5

```

DELETE and PATCH Method

200 OK



DELETE <http://127.0.0.1:5000/api/v1/students/2>

Auth type: No Auth

Key	Value	Description
Cache-Control	no-cache	
Postman-Token	<calculated when request is sent>	
Host	<calculated when request is sent>	
User-Agent	PostmanRuntime/2.5.0	
Accept	*/*	
Accept-Encoding	gzip, deflate, br	
Connection	keep-alive	
Authorization	((bearer_token_0nhe))	

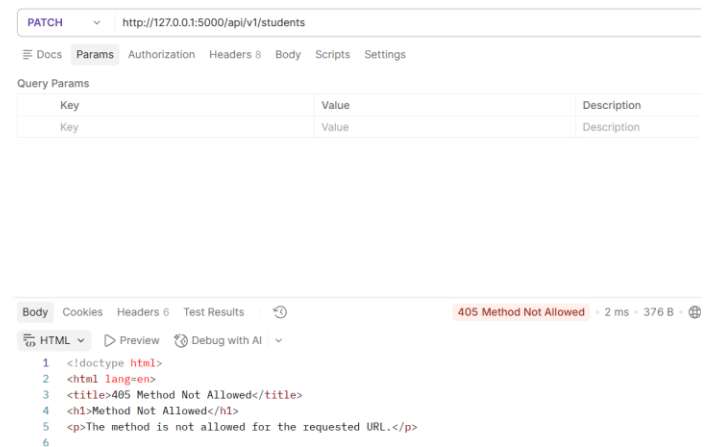
Body: 200 OK - 4 ms - 229 B

```

1 {
2   "message": "Student with ID 2 deleted",
3   "success": true
4 }
5

```

405 Method Not Allowed



PATCH <http://127.0.0.1:5000/api/v1/students>

Auth type: No Auth

Key	Value	Description
Key	Value	Description

Body: 405 Method Not Allowed - 2 ms - 376 B

```

1 <!doctype html>
2 <html lang=en>
3 <title>405 Method Not Allowed</title>
4 <h1>Method Not Allowed</h1>
5 <p>The method is not allowed for the requested URL.</p>
6

```

Learning outcomes (What I have learnt):

1. Understood the basics of RESTful web services and API testing.
2. Created and executed GET, POST, PUT, and DELETE requests using Postman.
3. Verified HTTP status codes, headers, and response payloads.
4. Validated API responses against expected results.

Evaluation Grid:

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Demonstration and Performance		12
2.	Worksheet		8
3.	Viva		10